

EduGenie-AI

Project Planning Phase

Team ID: 49A11A5ACEDC157AB004AFE76C371D3D

Project Type: AI / Generative AI Powered Learning Assistant

1. Project Overview

EduGenie-AI is an AI-powered learning assistant designed to help students learn more effectively. The application uses Generative AI capabilities to provide learning support, explanations, question answering, and personalized educational assistance.

2. Problem Statement

Students often need quick explanations, study support, practice questions, and learning resources outside the classroom. Traditional learning methods may not provide immediate and personalized assistance. EduGenie-AI aims to provide a single, user-friendly platform for accessible AI-based learning support.

3. Project Objectives

- Provide students with an AI-based learning assistant.
- Give simple and understandable explanations for academic topics.
- Support question answering and interactive learning.
- Reduce the time required to search for learning support.
- Create a simple and student-friendly interface.
- Demonstrate the practical use of Generative AI in education.

4. Proposed Features

Module	Planned Function
AI Learning Assistant	Answers student questions and provides explanations.
Topic Assistance	Helps students understand selected academic topics.
Interactive Q&A	Allows students to ask questions and receive responses.
Learning Support	Provides useful study-oriented guidance and content.
User Interface	Simple interface designed for easy student interaction.
AI Integration	Uses Generative AI to produce contextual responses.

5. Technology Stack

Area	Technology / Tool
Programming	Python
Backend	FastAPI / Uvicorn
AI / Generative AI	Google Gemini API / Generative AI
Frontend	HTML, CSS, JavaScript (as applicable)
Development Environment	Visual Studio Code
Version Control	Git / GitHub

6. Major Project Modules

1. **User Interface Module:** Provides the screens through which students interact with EduGenie.
2. **Input / Query Module:** Accepts questions or learning requests from the user.
3. **AI Processing Module:** Sends the user request to the Generative AI service and processes the response.
4. **Response Module:** Displays the generated answer in a clear and readable format.
5. **Testing Module:** Checks functional behaviour, user input, responses, and interface usability.

7. Project Development Plan

Phase	Activities	Status / Target
1. Planning	Problem identification, objectives, requirements	Completed / Planned
2. Design	UI design, module design, workflow	Planned
3. Development	Frontend, backend and AI integration	Planned / In Progress
4. Integration	Connect application components and AI service	Planned
5. Testing	Functional and user-interface testing	Planned
6. Documentation	Prepare reports, screenshots and presentation	Planned
7. Deployment / Demo	Run application and demonstrate features	Planned

8. Team Responsibilities

- **Team Leader:** Coordinate project activities, track progress, and support integration.
- **Development Team:** Implement frontend, backend, and AI-related components.
- **Testing & Documentation:** Verify features, record test results, and prepare project documentation.
- **All Members:** Participate in research, development, testing, presentation, and final demonstration.

9. Functional Requirements

- The system should accept user learning queries.
- The system should process queries through the configured AI service.
- The system should display relevant generated responses.
- The interface should allow users to interact easily with the assistant.
- The application should handle invalid or empty input appropriately.

10. Non-Functional Requirements

- **Usability:** The application should be simple and student-friendly.
- **Performance:** Responses should be delivered within a reasonable time.
- **Reliability:** The application should handle common errors gracefully.
- **Maintainability:** Code and modules should be organized for future improvements.
- **Security:** API keys and sensitive configuration should not be exposed in the user interface or source repository.

11. Expected Outcome

The completed EduGenie-AI application is expected to provide students with an easy-to-use AI learning assistant. The project will demonstrate how Generative AI can be integrated with a web application to support interactive and personalized learning.

12. Future Enhancements

- Personalized learning recommendations.
- Quiz and practice-question generation.
- Progress tracking and learning history.
- Support for additional subjects and languages.
- Voice-based interaction.
- Improved personalization using student learning preferences.

Conclusion

EduGenie-AI is planned as a practical educational application that combines Generative AI with a student-focused interface. The planning phase establishes the requirements, modules, technologies, development stages, and expected outcomes needed to guide the project toward implementation and testing.

